

Threats to identification

Slide route: Lecture8v2.tex: external/internal validity, measurement error, missing data, selection, simultaneity, SEs and prediction. Chapter 9.

Session 11 | Diagnose before choosing a remedy

Outcomes: Separate internal from external validity; explain classical attenuation, selection, and simultaneity; match threats to plausible responses.

100-minute route

0–10	Post-midterm retrieval: interpretation versus identification.
10–25	External and internal validity; discuss slide examples.
25–45	OVB, wrong functional form, measurement error.
45–55	Pause and attenuation calculation.
55–75	Missing data, selection, simultaneity.
75–90	Inconsistent SEs, validity checklist, prediction.
90–100	Exit diagnosis and bridge to IV.

Slide-by-slide script

Slide 1 | Title

Guide Open by asking why correct OLS arithmetic and standard errors may still leave a causal claim unsupported.

Slide 2 | Roadmap

Guide Preview external validity, internal threats, measurement, missing data, selection, simultaneity, inference, and prediction.

Slide 3 | Last lesson

Guide Retrieve the interaction effect at a stated value of the interacting variable.

Slide 4 | Last lesson

Guide Ask which assumptions concern identification and which concern inference.

Slide 5 | Classic pitfalls

Guide Tell students that a threat is a mechanism that changes the estimand or invalidates its interpretation.

Slide 6 | External validity

Guide Distinguish transportability to other people, places, and times from internal validity in the study sample.

Slide 7 | Meta-analysis

Guide Use the study examples to ask what population the evidence is meant to describe.

Slide 8 | Meta-analysis

Guide Ask whether a larger collection of studies automatically removes publication or selection problems.

Slide 9 | Internal validity

Guide Define internal validity as credibility of the causal comparison for the studied population.

Slide 10 | OVB revision

Guide Draw the omitted-variable path and ask whether adding controls solves a problem only when the controls are appropriate and measured.

Slide 11 | OVB revision

Guide Connect the threat to the sign formula from Class 7.

Slide 12 | Errors-in-variables

Guide Write $X = X^* + v$ and distinguish noise in a regressor from noise in the outcome.

Slide 13 | Errors-in-variables

Guide Ask whether attenuation is a finite-sample imprecision problem. No, it changes the probability limit in the classical setup.

Slide 14 | Classical measurement error

Guide Calculate $2 \times 9 / (9 + 3) = 1.5$ and explain the reliability ratio.

Slide 15 | Classical measurement error

Guide Qualify the toward-zero result: it requires the simple classical assumptions and does not generalize to every multiple-regression setting.

Slide 16 | Missing data

Guide Ask what determines whether dropping incomplete observations changes the target or creates selection.

Slide 17 | Missing data

Guide Distinguish missing completely at random, missing related to observed variables, and missing related to unobserved outcomes conceptually.

Slide 18 | Sample selection

Guide Use wages observed only among employed people. Ask whether the observed slope describes wage offers for everyone.

Slide 19 | Sample selection

Guide Explain why robust SEs do not repair selection into the sample.

Slide 20 | Simultaneity

Guide Use price and quantity: both respond to supply and demand shocks, so covariance need not identify either curve.

Slide 21 | Inconsistent standard errors

Guide Distinguish unequal variance, dependence, and endogeneity. Each requires a different response.

Slide 22 | Validity checklist

Guide Have students name the estimand, source of variation, threat, and assumption before choosing a remedy.

Slide 23 | Prediction

Guide End with the distinction between predictive stability and causal validity. Ask whether an endogenous predictor can still forecast well.

Worked example | Measurement error

Use with slides Lecture8v2 slides 12–15: Errors-in-variables and Classical Measurement Error. Work the attenuation calculation on slide 14 or 15, after students have seen why the measured regressor contains noise.

Suppose the true relationship is $Y = \beta_0 + 2X^* + u$, but we observe $X = X^* + v$. Under classical regressor measurement error, v is mean zero and uncorrelated with X^* and u . If $\text{Var}(X^*) = 9$ and

$\text{Var}(v) = 3$:

$$\text{plim } \hat{\beta}_{\text{OLS}} = 2 \frac{9}{9+3} = 1.5.$$

Ask Does a larger sample restore the coefficient to 2? No. It concentrates the estimator around the attenuated target. Better measurement or a valid instrument addresses a different problem from increasing precision.

Qualification Attenuation toward zero is a result for this simple classical setup; do not generalize its direction to arbitrary measurement error or every coefficient in multiple regression. Classical noise in the outcome alone need not bias slopes, though it raises uncertainty.

Cases for discussion | Threat, mechanism, response

Selection Wages are observed only among employed people. If employment depends on unobserved wage determinants, the observed sample need not represent the wage-offer relationship for everyone. More complete data or a justified selection design is needed; merely adding a robust SE is insufficient.

Simultaneity Price and quantity are jointly determined by demand and supply. Their observed covariance need not trace a demand curve. A supply shifter excluded from demand is a possible instrument, provided its exclusion and relevance can be defended.

External validity An internally valid tutoring experiment at one university need not identify the same effect for all schools. Ask what differs: students, implementation, context, or treatment intensity.

Dependence Repeated outcomes for one district can be correlated. Heteroskedasticity-robust SEs alone are insufficient; a dependence-aware variance estimator may be needed.

Pair activity | Classify three claims

1. A treatment improves outcomes in one city, so it must do so everywhere.
2. Self-reported education contains substantial recall error.
3. A high- R^2 demand regression uses equilibrium prices.

Answers: External validity; potential errors-in-variables; simultaneity despite fit. For each, ask what extra data or design evidence would make the claim more credible.

Missing data caution Dropping incomplete rows is not automatically harmless. Whether complete-case analysis is valid depends on why data are missing and the estimand. Avoid treating all missingness as one problem with one remedy.

Prediction distinction An endogenous variable can be predictive in a stable environment, yet policy interventions can change its relationship to the outcome. Evaluate predictions out of sample; evaluate causal claims through design and assumptions.

If short / preparation Use one external-validity illustration and three internal threats in depth. Read Chapter 12 and come prepared to propose an instrument and criticize its exclusion restriction.

Source: Lecture8v2.tex; numerical and discussion examples are illustrative.